

Ultrafast X-Ray Experiments

**Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21**

Léonie Lanz
Nico Kalogerakis

September 2, 2026

Contents

1. Einleitung	1
2. Theoretical Background	1
2.1. Synchrotronradiation	1
2.1.1. Free Electron Laser (FEL)	1
2.1.2. Undulators	1
2.2. The investigated Molecule	1
2.2.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$	1
2.2.2. Photophysics	1
2.3. Pump-Probe-Principle	1
2.3.1. The Pump-Pulse	1
2.3.2. The Probe-Pulse	1
3. Experimental Setup	1
3.1. Software: VLab	1
3.2. Undulator and Beamline	1
3.3. Monochromator and Focusing Optics	1
3.4. Spectrum Analyser	1
3.5. Time-arrival Detector	1
3.6. Sample Environment	1
3.7. Laser System	1
3.8. Von Hamos Spectrometer	1
4. Experimental Procedure	1
4.1. Preparation of the Experiment	1
4.1.1. Alignment and Characterization of the X-ray Beam	1
4.1.2. Preparatory Estimates for the Experiment	2
4.2. Performing the Experiment with Data Acquisition	2
5. Analysis and Interpretation	2
6. Discussion of the physical meaning	2
7. Conclusion and Outlook	2
References	A
Abbildungsverzeichnis	B
Appendix A. Tabellen	C
Appendix B. Code	C

1. Einleitung

2. Theoretical Background

2.1. Synchrotronradiation

2.1.1. Free Electron Laser (FEL)

2.1.2. Undulators

2.2. The investigated Molecule

2.2.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$

2.2.2. Photophysics

2.3. Pump-Probe-Principle

2.3.1. The Pump-Pulse

2.3.2. The Probe-Pulse

3. Experimental Setup

3.1. Software: VLab

Datenstruktur (beam imaging and xray eye): 420x420 pixel, bei biu screen dim von 10mm x10mm, bei xray eye screen 2mm x 2mm

3.2. Undulator and Beamline

3.3. Monochromator and Focusing Optics

monochromator focusing optics: mirrors, power slits, solid attenuators, refractive Lenses, X-ray eye

3.4. Spectrum Analyser

diffraction gratings

3.5. Time-arrival Detector

3.6. Sample Environment

3.7. Laser System

3.8. Von Hamos Spectrometer

Quelle Versuchsmappe:[1]

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.

As the absorption edge for the Fe $K\beta_{1,3}$ line of 7065 eV, the energy of the xray beam is set to 7400 eV, to avoid the ripples near the reaction (siehe ??) but without too strong of a loss in intensity. To measure

the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturation of the beam imaging unit 2 (BIU2):

The size and position of the x-ray is now measured using BIU2. Now the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24° .

For the following experiment the monochromatic beam is used as it leads to a higher resolution than the pink x-ray, even though the intensity of the beam suffers.

The last step of diagnosing the beam line is using the refractive beryllium x-ray lenses to focus the beam at the imaging unit x-ray eye. It's important to appropriately adjust the attenuator foils as to not burn the imaging unit with a focussed beam.

With beryllium lenses 1, 2 and 5 (taken from ??) and the attenuator foils listed here ??, a beam with the FWHM of 13 μm (siehe ??) is achieved. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [?] and the properties of the lenses themselves taken from ??:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$

$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4,76 \text{ m}$$

—Notes—

X-ray energy $E_0 = 7400 \text{ eV}$ (has to be a good bit away from reaction edge of 7065 keV to be stable)

Alignment: attenuators to not burn through out screens (here: 4: B4C 0.8, 3: C 0.4, 2: Si 0.05, C 0.1) size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

optical resolution: 25 $\mu\text{m}/\text{px}$

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

Fun Facts: molar mass: $M([Fe(bipy)_3]^{2+}) \approx M([Fe(bipy)_3]) = 524,409 \frac{\text{g}}{\text{mol}}$ concentration of the sample: $c = 0,380 \frac{\text{mol}}{\text{L}}$ total attenuation of X-rays: attenuation comparison plot flow speed of the jet: $v_{jet} = 11 \frac{\text{m}}{\text{s}}$

Laser: excitation wavelength: $\lambda_{pump} = 400 \text{ nm}$ (dazu alle fun facts mit MLCT Zuständen und so, und nicht overshooten lol) focussing, diameter of the focus Overfilling: beam sizes of the optical laser and the X-ray pulse relative to each other (plot beam diameter)

Spectrometer: Van Hamos Spectrometer vs Johann Spectrometer (plot of the two geometries) crystal selection: $Si(531)$ vs $Ge(531)$ (plot of the two crystal reflections)

4.2. Performing the Experiment with Data Acquisition

5. Analysis and Interpretation

6. Discussion of the physical meaning

7. Conclusion and Outlook

References

- [1] C. Bressle, “Ultrafast x-ray experiments virtual experiments at the x-ray laser european xfel.”

List of Figures

Appendix A Tabellen

Appendix B Code